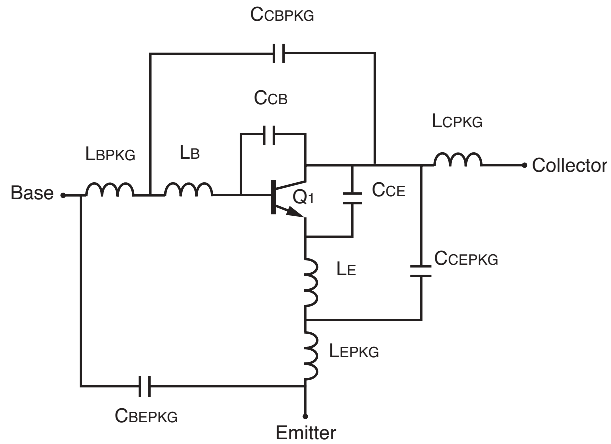
CEL

NONLINEAR MODEL NESG2031M05

SCHEMATIC



BJT NONLINEAR MODEL PARAMETERS[1]

Parameters	Q1	Parameters	Q1
IS	1.593e-15	MJC	0.125
BF	305.3	XCJC	1
NF	1.099	CJS	0
VAF	35.92	VJS	0.75
IKF	94.86e-3	MJS	0
ISE	27.45e-15	FC	0.8
NE	2.035	TF	4e-12
BR	21.10	XTF	10
NR	1.066	VTF	5
VAR	2.782	ITF	0.5
IKR	20.46e-3	PTF	20
ISC	11.77e-18	TR	0
NC	2.0	EG	1.11
RE	1.6	XTB	1.3
RB	1.0	XTI	5.2
RBM	50e-3	KF	0
IRB	1e-4	AF	1
RC	7.2		
CJE	642e-15		
VJE	751e-3		
MJE	93.06e-3		
CJC	163.4e-15		
VJC	0.56		

(1) Gummel-Poon Model

ADDITIONAL PARAMETERS

Parameters	NESG2031M05
C_{CB}	0.01 pF
C_{CE}	0.07 pF
L_B	0.25 nH
L_E	0.15 nH
C_{CBPKG}	0.13 pF
C_{CEPKG}	0.01 pF
C_{BEPKG}	0.03 pF
L_{BPKG}	0.9 nH
L_{CPKG}	1.0 nH
L_{EPKG}	0.19 nH

MODEL TEST CONDITIONS
Frequency: 0.1 to 6 GHz
Bias: V_{CE} = 2 V, I_C = 1 mA to 20 mA
Date: 09/2003

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